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FRACTIONAL CAUCHY PROBLEMS ASSOCIATED WITH THE BI-ORDINAL HILFER FRACTIONAL q-DERIVATIVE

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Abstract. The aim of this paper is the investigation of the existence and uniqueness of solutions to Cauchy-type problems for fractional q-difference equations with the bi-ordinal Hilfer fractional q-derivative which is an extension of the Hilfer fractional q-derivative. An approach is based on the equivalence of the nonlinear Cauchy-type problem with a nonlinear Volterra q-integral equation of the second kind. Applying an analog of Banach's fixed point theorem we prove the uniqueness and the existence of the solution. Moreover, we present an explicit solution to the q-analog of the Cauchy problem for the linear case.

1. Introduction

The quantum groups provide the key to q-deforming the basic structures of physics from the point of view of non-commutative geometry. The theory plays an important role in the conformal field theory, exact soluble models in statistical physics [7] and in a wide range of applications, from cosmic strings and black holes to the solid state physics problems [5], [8], [25], [33]. For instance, the q-deformed Lie algebras are extended versions of the usual Lie algebras which are beyond the scope of usual Lie algebras [8]. Moreover, it has recently led to the application of the q-calculus in the construction of generalized statistical mechanics where non-extensivity properties can arise from the q-deformed theory [16]. For some analytical constructions on quantum groups, we refer to [6].

Recently, mathematicians have paid much attention to the q-calculus [11], [12], [13] and fractional q-differential equations [1], [2], [3], [26], [28], [27], [29]. The q-calculus (or quantum calculus) can be dated back to the beginning of the twentieth century, to F. H. Jackson and R. D. Carmichael's work [10], [19], [20]. There have been some papers dealing with the existence and uniqueness or multiplicity of solutions for nonlinear fractional q-difference equations by the use of some well-known fixed point theorems [24]. For some recent developments on the subject, see e.g. [4], [14], [15], [32] and the references therein.

The notations used in this introduction are explained in Section 2 below. In this paper, we also focus more on the q-analogue of Hilfer's bi-ordinal fractional derivative

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or composite fractional derivative operator (see Definition 3.1) and we derive sufficient conditions for the existence of a unique solution of the Cauchy-type q-fractional problem (3.3)-(3.4). Moreover, for the proof of this theorem, we prove an equivalence theorem (Theorem 3.4) of independent interest. In the case when $\mu = 0$ this is the generalized Riemann–Liouville fractional q-derivative (see (2.3)) and in case when $\mu = 1$ it corresponds to the Caputo fractional q-derivative (see [27]).

The paper is organized as follows: The main results are presented and proved in Section 3 and the announced examples are given in Section 4. In the final section, we present the explicit solution to the Cauchy problem for the linear case, applying the successive-iteration method. In order to facilitate presentations, we include in Section 2 some necessary preliminaries.

2. Preliminaries

First, we recall several concepts of the q-calculus that we need in the paper. We assume that $0 < q < 1$.

Let $\alpha \in \mathbb{R}$. Then a q-real number $[\alpha]_q$ is defined by ([11])

$$[\alpha]_q := \frac{1-q^\alpha}{1-q},$$

and the q-binomial coefficient $[n]_q!$ is defined by ([11])

$$[n]_q! := \begin{cases} 1, & \text{if } n = 0, \\ [1]_q \times [2]_q \times \cdots \times [n]_q, & \text{if } n \in \mathbb{N}. \end{cases}$$

The q-shifted operations for $n \in \mathbb{N}$ are defined by

$$(a;q)_0 = 1, \ (a;q)_n = \prod_{k=0}^{n-1} \left(1 - q^k a\right), \ (a;q)_\infty = \lim_{n\to\infty} (a;q)_n.$$

Moreover,

$$(a;q)_\alpha = \frac{(a;q)_\infty}{(q^\alpha a;q)_\infty}. \tag{2.1}$$

The q-gamma function $\Gamma_q(x)$ is defined by ([11])

$$\Gamma_q(x) = \frac{(q;q)_\infty}{(q^x;q)_\infty}(1-q)^{1-x} \tag{2.2}$$

for any $x > 0$. This q-Gamma function has the following property ([11]):

$$\Gamma_q(x)[x]_q = \Gamma_q(x+1).$$

The q-derivative of a function f with respect to x is defined by ([19])

$$D_q f(x) = \frac{f(x) - f(qx)}{x(1-q)},$$

and the q-derivatives $D_q^n(f(x))$ of higher order are defined inductively as follows ([26]):

$$D_q^0(f(x)) := f(x), \quad D_q^n(f(x)) := D_q\left(D_q^{n-1} f(x)\right), (n = 1, 2, 3, \dots).$$

The general q-integral (or Jackson integral) is defined ([20]) as

$$\int_a^b f(x)d_qx = \int_0^b f(x)d_qx - \int_0^a f(x)d_qx,$$

for $0 < a < b$, where

$$\int_0^a f(x)d_qx = (1-q)a\sum_{m=0}^{\infty} q^m f(aq^m).$$

In the middle of the last century, W. A. Al-Salam [2] and R. P. Agarwal [3] presented a q-analog of the Riemann–Liouville fractional integral with only zero as a lower limit of the q-integration and a fractional q-derivative. After that, P. M. Rajkovic', S. D. Marinkovic' and M. S. Stankovic' [26] (see also [27]) allowed the lower limit of the q-integration to be nonzero and reintroduced the Riemann-Liouville q-fractional integrals $I^{\alpha}_{q,a+}f$ of order $\alpha > 0$ by

$$\left(I^{\alpha}_{q,a+}f\right)(x) = \frac{x^{\alpha-1}}{\Gamma_q(\alpha)}\int_a^x (qt/x;q)_{\alpha-1}f(t)d_qt. \tag{2.3}$$

The Riemann-Liouville fractional q-derivative $D^{\alpha}_{q,a+}f$ of order $\alpha > 0$ is defined by

$$\left(D^{\alpha}_{q,a+}f\right)(x) = \left(D^{[\alpha]}_{q,a+}I^{[\alpha]-\alpha}_{q,a+}f\right)(x),$$

where $[\alpha]$ denotes the smallest integer greater or equal to α.

For $\alpha > 0$ and $\lambda > -1$ the following identities hold [27, Lemma 12]:

$$\left(I^{\alpha}_{q,a+}t^{\lambda}(t/a;q)_{\lambda}\right)(x) = \frac{\Gamma_q(\lambda+1)}{\Gamma_q(\alpha+\lambda+1)}x^{\alpha+\lambda}(x/a;q)_{\alpha+\lambda} \tag{2.4}$$

and

$$\left(D^{\alpha}_{q,a+}t^{\lambda}(t/a;q)_{\lambda}\right)(x) = \frac{\Gamma_q(\lambda+1)}{\Gamma_q(\lambda+1-\alpha)}x^{\lambda-\alpha}(x/a;q)_{\lambda-\alpha}, \quad \alpha \le \lambda, \tag{2.5}$$

and

$$\left(D^{\alpha}_{q,a+}t^{\lambda}(t/a;q)_{\lambda}\right)(x) = 0, \quad \alpha > \lambda. \tag{2.6}$$

For all further discussions, we assume that the functions are defined on an interval $[0,b]$ and $b > 0$. Moreover, $a \in [0,b]$ is an arbitrary fixed point. In [4, Subsection 4.3], M. H. Annaby and Z. S. Mansour presented definitions of spaces of q-integrable, q-absolutely continuous on $[0,a]$ functions. We also give characterizations of those modified spaces $[a,b]$ which will be used later. For $1 \le p < \infty$ we define the space $L^p_q = L^p_q[a,b]$ by

$$L^p_q[a,b] := \left\{ f : [a,b] \to \mathbb{R} : \left(\int_a^b |f(x)|^p d_qx\right)^{\frac{1}{p}} < \infty \right\}.$$

Let $\alpha > 0$, $\beta > 0$ and $1 \leq p < \infty$. Then the q–fractional integrals have the following semi-group property (see [28, Lemma 2.3]):

$$\left(I^{\alpha}_{q,a+}I^{\beta}_{q,a+}f\right)(x) = \left(I^{\alpha+\beta}_{q,a+}f\right)(x), \tag{2.7}$$

for all $x \in [a,b]$ and $f \in L^p_q[a,b]$.

Let $\alpha > \beta > 0$, $1 \leq p < \infty$ and $f \in L^p_q[a,b]$. Then the following (see [28, Lemma 2.3]) equalities

$$\left(D^{\alpha}_{q,a+}I^{\alpha}_{q,a+}\right)(x) = f(x), \quad \left(D^{\beta}_{q,a+}I^{\alpha}_{q,a+}f\right)(x) = \left(I^{\alpha-\beta}_{q,a+}f\right)(x), \tag{2.8}$$

hold for all $x \in [a,b]$.

A function $f : [a,b] \to \mathbb{R}$ is called q-absolutely continuous if $\exists \varphi \in L^1_q[a,b]$ such that

$$f(x) = f(a) + \int_a^x \varphi(t) d_q t$$

for all $x \in [a,b]$ (see also, [28, Definition 2.4]).

The collection of all q-absolutely continuous functions on $[a,b]$ is denoted by $AC_q[a,b]$. For $n = 1,2,3,\ldots$ we denote by $AC^n_q[a,b]$ the space of real-valued functions $f(x)$ which have q-derivatives up to order $n-1$ on $[a,b]$ such that $D^{n-1}_q f \in AC_q[a,b]$:

$$AC^n_q[a,b] := \left\{f : [a,b] \to \mathbb{R}; D^{n-1}_q f \in AC_q[a,b]\right\}.$$

The function $f : [a,b] \to \mathbb{R}$ belongs to $AC^n_q[a,b]$ if the following equality holds (see [28, Lemma 2.5]):

$$f(x) = \frac{x^{n-1}}{\Gamma_q(n)} \int_a^x (qt/x;q)_{n-1} \varphi(t) d_q t + \sum_{k=0}^{n-1} c_k x^k (a/x;q)_k,$$

where $\varphi(x) := D^n_q f(x)$ and $c_k = \frac{D^k_q f(a)}{\Gamma_q(k)}$, $k = 0,1,2,\ldots,n-1$, are arbitrary constants.

Let $f \in L^1_q[a,b]$ and $\left(I^{n-\alpha}_{q,a+}f\right) \in AC^n_q[a,b]$ with $n = [\alpha], \alpha > 0$. Then the following equality holds (see [28, Lemma 2.5]):

$$\left(I^{\alpha}_{q,a+}D^{\alpha}_{q,a+}f\right)(x) = f(x) - \sum_{k=1}^{n} \frac{\left(D^{\alpha-k}_{q,a+}f\right)(a)}{\Gamma_q(\alpha-k+1)} x^{\alpha-k}(a/x;q)_{\alpha-k}. \tag{2.9}$$

Let $\alpha > 0$ and $1 \leq p < \infty$. Then the fractional integration operator $I^{\alpha}_{q,a+}$ is bounded in $L^p_q[a,b]$ (see [28, Lemma 2.6]):

$$\|I^{\alpha}_{q,a+}f\|_{L^p_q[a,b]} \leq K\|f\|_{L^p_q[a,b]}, \tag{2.10}$$

where $K = \frac{(b-qa)^{\alpha}_q}{\Gamma_q(\alpha+1)}$ and $(b-qa)^{\alpha}_q = b^{\alpha}(qa/b;q)_{\alpha}$.

3. Main results

Now, we start to present the concept of the bi-ordinal fractional q-derivative generalizing the well-known Hilfer's q-derivative (see [17], [18] and [30]). Bi-ordinal Hilfer derivative was introduced for the first time by Bulavatsky in [9] in the case when the fractional order lies between 0 and 1. Later, in [21] it was generalized for higher fractional orders. Boundary-value problems for PDEs involving this fractional derivative were considered in recent works [22] and [31].

Below for the first time, we introduce q-analogue of this integral-differential operator together with the key property which allows us to solve the Cauchy problem for the q-fractional differential equation.

Definition 3.1. We define the bi-ordinal Hilfer fractional q-derivative $\mathcal{D}_{q,a+}^{(\alpha,\beta)\mu}$ of orders α and β, and type μ with respect to x by

$$\left(\mathcal{D}_{q,a+}^{(\alpha,\beta)\mu} f\right)(x) := \left(I_{q,a+}^{\mu(1-\alpha)} D_q \left(I_{q,a+}^{(1-\mu)(1-\beta)} f\right)\right)(x), \tag{3.1}$$

for $0 < \alpha, \beta \leq 1$ and $0 \leq \mu \leq 1$.

In the case when $\mu = 0$, the generalized fractional q-derivative (3.1) would correspond to the the Riemann-Liouville fractional q-derivative of order β (see (2.3)) and in the case when $\mu = 1$ it corresponds to the Caputo fractional q-derivative $\left({}_cD_{q,a+}^{\alpha} f\right)(x)$ defined by (see [27]):

$$\left({}_cD_{q,a+}^{\alpha} f\right)(x) := \left(I_{q,a+}^{1-\alpha} D_q f\right)(x) = \frac{x^{-\alpha}}{\Gamma_q(1-\alpha)} \int_a^x (qt/x;q)_{-\alpha} f(t) d_q t.$$

Remark 3.2. Let $\gamma = \beta + \mu(n-\beta)$ and $\nu = \beta + \mu(\alpha - \beta)$ for $n-1 < \alpha, \beta \leq n, n \in \mathbb{N}$ and $0 \leq \mu \leq 1$. The generalized fractional q-derivative $D_{q,a+}^{(\alpha,\beta)\mu} f$ can be represented as follows:

$$\begin{aligned} \left(D_{q,a+}^{(\alpha,\beta)\mu} f\right)(x) &:= \left(I_{q,a+}^{\mu(n-\alpha)} D_q^n \left(I_{q,a+}^{(1-\mu)(n-\beta)} f\right)\right)(x) \\ &= \left(I_{q,a+}^{\gamma-\nu} D_q^n \left(I_{q,a+}^{n-\gamma} f\right)\right)(x) \\ &= \left(I_{q,a+}^{\gamma-\nu} D_{q,a+}^{\gamma} f\right)(x). \end{aligned} \tag{3.2}$$

Next, we study the question of the equivalence between the following Cauchy-type q-fractional problem

$$\left(D_{q,a+}^{(\alpha,\beta)\mu} y\right)(x) = f(x, y(x)), \quad n-1 < \alpha \leq n; n \in \mathbb{N}, 0 \leq \beta \leq 1, \tag{3.3}$$

$$\lim_{x \to a+} \left(D_q^k I_{q,a+}^{(1-\mu)(n-\beta)} y\right)(x) = \xi_k, \quad \xi_k \in \mathbb{R}, k = 1, 2, \ldots n, \tag{3.4}$$

and the Volterra q-integral equation of the second kind:

$$y(x) = \sum_{k=1}^{n} \frac{\xi_k}{\Gamma_q(\gamma - k + 1)} x^{\gamma-k} (a/x;q)_{\gamma-k} + \left(I_{q,a+}^{\nu} f(\cdot, y(\cdot))\right)(x). \tag{3.5}$$

Theorem 3.3. *We assume $n-1 < \alpha, \beta \leq n, n \in \mathbb{N}$, $0 \leq \mu \leq 1$, $\gamma = (n-\alpha)(1-\beta)$, $\nu = \beta + \mu(\alpha - \beta)$. Let $f(\cdot,\cdot) : [a,b] \times \mathbb{R} \to \mathbb{R}$ be a function such that $f(\cdot, y(\cdot)) \in L_q^1[a,b]$*

for all $y \in L_q^1[a,b]$. Then y satisfies the relations (3.3)-(3.4) if and only if y satisfies the integral equation (3.5).

Proof. We assume that $y \in L_q^1[a,b]$ satisfies the relations (3.3)-(3.4). Since $f(x,y(x)) \in L_q^1[a,b]$ we get that $\left(D_{q,a+}^{(\alpha,\beta)\mu} y\right) \in L_q^1[a,b]$ for $\gamma = \beta + \mu(n-\beta)$, $\nu = \beta + \mu(\alpha - \beta)$ together with $n-1 < \alpha, \beta \leq n, n \in \mathbb{N}$, $0 \leq \mu \leq 1$. According to (2.10) we obtain that $\left(I_{q,a+}^{\nu} f\right) \in L_q^1[a,b]$. Applying the integral operator $I_{q,a+}^{\nu}$ to both sides of (3.3) we obtain

$$I_{q,a+}^{\nu} D_{q,a+}^{(\alpha,\beta)\mu} y(x) = \left(I_{q,a+}^{\nu} f(\cdot, y(\cdot))\right)(x). \tag{3.6}$$

From Remark 3.2 and (2.7), (2.9) it follows that

$$\begin{aligned}
\left(I_{q,a+}^{\nu} D_{q,a+}^{(\alpha,\beta)\mu} y\right)(x) &= \left(I_{q,a+}^{\nu} I_{q,a+}^{\gamma-\nu} D_{q,a+}^{\gamma} y\right)(x) \\
&= \left(I_{q,a+}^{\gamma} D_{q,a+}^{\gamma} y\right)(x) \\
&= y(x) - \sum_{k=1}^{n} \frac{\left(D_{q,a+}^{\gamma-k} f\right)(a)}{\Gamma_q(\gamma-k+1)} x^{\gamma-k} (a/x;q)_{\gamma-k} \\
&= y(x) - \sum_{k=1}^{n} \frac{\left(D_q^k I_{q,a+}^{(1-\mu)(n-\beta)} f\right)(a)}{\Gamma_q(\gamma-k+1)} x^{\gamma-k} (a/x;q)_{\gamma-k}. \qquad (3.7)
\end{aligned}$$

According to (3.4), (3.6) and (3.7) we get the equation (3.5). The necessity is proved.

Now we prove the sufficiency. Let $y \in L_q^1[a,b]$ satisfy equation (3.5). Applying the operator $D_{q,a+}^{(\alpha,\beta)\mu}$ to both sides of (3.5), we have that

$$\begin{aligned}
\left(D_{q,a+}^{(\alpha,\beta)\mu} y\right)(x) &= \sum_{k=1}^{n} \frac{\xi_k}{\Gamma_q(\gamma-k+1)} \left(D_{q,a+}^{(\alpha,\beta)\mu} t^{\gamma-k} (a/t;q)_{\gamma-k}\right)(x) \\
&+ D_{q,a+}^{(\alpha,\beta)\mu} \left(I_{q,a+}^{\nu} f(\cdot, y(\cdot))\right)(x). \qquad (3.8)
\end{aligned}$$

From (2.6) it follows that

$$\left(D_{q,a+}^{\gamma} t^{\gamma-k} (a/t;q)_{\gamma-k}\right)(x) = 0, \quad 1 \leq k \leq n. \tag{3.9}$$

By Remark 3.2 and (2.7), (2.8) and (3.9), we find that

$$\left(D_{q,a+}^{(\alpha,\beta)\mu} t^{k-\gamma} (a/t;q)_{k-\gamma}\right)(x) = \left(I_{q,a+}^{\gamma-\nu} D_{q,a+}^{\gamma} t^{\gamma-k} (a/t;q)_{\gamma-k}\right)(x) = 0$$

and using (3.2) we get that

$$\begin{aligned}
\left(D_{q,a+}^{(\alpha,\beta)\mu} I_{q,a+}^{\nu} f\right)(x) &= \left(I_{q,a+}^{\gamma-\nu} D_q^{\gamma} I_{q,a+}^{\nu} f(\cdot, y(\cdot))\right)(x) \\
&= \left(I_{q,a+}^{\gamma-\nu} I_{q,a+}^{\nu-\gamma} f(\cdot, y(\cdot))\right)(x) \\
&= f(x, y(x)).
\end{aligned}$$

Therefore, we can rewrite (3.8) in the form

$$\left(D_{q,a+}^{(\alpha,\beta)\mu} y\right)(x) = f(x, y(x)).$$

Finally, we will show that the initial condition of (3.4) also holds. For this aim, we apply the operator $I_{q,a+}^{(1-\mu)(n-\beta)}$ to both sides of (3.5) and using (2.4), (2.7), we conclude that

$$\begin{aligned}\left(I_{q,a+}^{(1-\mu)(n-\beta)}y\right)(x) &= \sum_{k=1}^{n}\frac{\xi_k}{\Gamma_q(\gamma-k+1)}\left(I_{q,a+}^{(1-\mu)(n-\beta)}t^{\gamma-k}(a/t;q)_{\gamma-k}\right)(x)\\ &+ \left[I_{q,a+}^{(1-\mu)(n-\beta)}I_{q,a+}^{\nu}f(\cdot,y(\cdot))\right](x)\\ &= \sum_{k=1}^{n}\frac{\xi_k}{[n-k]_q!}x^{n-k}(a/x;q)_{n-k}+\left(I_{q,a+}^{\mu(\alpha-n)-n}f(\cdot,y(\cdot))\right)(x). \qquad (3.10)\end{aligned}$$

Let $0\le m\le n-1$. Then, we apply the operator D_q^m to both sides of (3.10) by using (2.5) and (2.9), to obtain that

$$\begin{aligned}\left(D_q^m I_{q,a+}^{(1-\mu)(n-\beta)}y\right)(x) &= \sum_{k=1}^{n}\frac{\xi_k}{[n-k]_q!}D_q^m\left[x^{n-k}(a/x;q)_{n-k}\right]+\left(D_q^m I_{q,a+}^{\mu(\alpha-n)-n}f(\cdot,y(\cdot))\right)(x)\\ &= \sum_{k=1}^{n}\frac{\xi_k}{[n-k-m]_q!}x^{n-k-m}(a/x;q)_{n-k-m}\\ &+ \left(I_{q,a+}^{\mu(\alpha-n)-n-m}f(\cdot,y(\cdot))\right)(x). \qquad (3.11)\end{aligned}$$

Taking in (3.11) the limit $x\to a+$, we get the relations in (3.4). Thus also the sufficiency is proved, which completes the proof of Theorem 3.3. □

In the next theorem, we give conditions for a unique global solution to the Cauchy-type problem (3.3)-(3.4) in the space $L^1_{\alpha,\beta,\mu,q}[a,b]$ defined for $\alpha>0$ by

$$L^1_{\alpha,\beta,\mu,q}[a,b]:=\left\{y\in L_q^1[a,b]: D_{q,a+}^{(\alpha,\beta)\mu}y\in L_q^1[a,b]\right\}.$$

The proof of the following existence and uniqueness theorem depends heavily on Theorem 3.3 and the Banach's fixed point Theorem (see [24, Subsection 8.1]).

Theorem 3.4. *We assume $\gamma=\beta+\mu(n-\beta)$ and $\nu=\beta+\mu(\alpha-\beta)$ for $n-1<\alpha,\beta\le n, n\in\mathbb{N}$ and $0\le\mu\le 1$. Let $n-1<\alpha\le n; n\in\mathbb{N}$, and let $f:[a,b]\times\mathbb{R}\to\mathbb{R}$ be a function such that $f(\cdot,v(\cdot))\in L_q^1[a,b]$ for all $v\in L_q^1[a,b]$, and it satisfies the Lipschitz condition in the following form:*

$$|f(x,v_1(x))-f(x,v_2(x))|\le A\,|v_1(x)-v_2(x)|, \qquad (3.12)$$

where $A>0$ does not depend on $x\in[a,b]$ and $v_1,v_2\in L_q^1[a,b]$. Then there exists a unique solution $y\in L^1_{\alpha,\beta,\mu,q}[a,b]$ to the Cauchy-type problem (3.3)-(3.4).

Proof. First we show that there exists a unique solution $y\in L_q^1[a,b]$. In view of Theorem 3.3 it is sufficient to prove the existence of a unique solution $y\in L_q^1[a,b]$ of the nonlinear Volterra q-integral equation (3.5). We assume that $[a,a_1]\subset[a,b]$ are such that

$$\omega_1:=AK\frac{(a_1-qa)_q^{\alpha}}{\Gamma_q(\alpha+1)}<1, \qquad (3.13)$$

where K is as in (2.10). Obviously, the equation (3.5) holds on the interval $[a, a_1]$. Consequently, we rewrite this equation in the form $y = Ty$ with

$$(Ty)(x) := y_0 + \left(I^{\nu}_{q,a+} f(\cdot, y(\cdot))\right)(x), \tag{3.14}$$

where $y_0 := \sum_{k=1}^{n} \frac{\xi_k}{\Gamma_q(\gamma-k+1)} x^{\gamma-k} (a/x; q)_{\gamma-k}$.

If $y \in L^1_q[a, a_1]$, then $Ty \in L^1_q[a, a_1]$. Moreover, for any $y_1, y_2 \in L^1_q[a, a_1]$, $f(\cdot, y_j(\cdot)) \in L^1_q[a, a_1]$, $j = 1, 2$, and using (2.10), (3.12) and (3.13) we have that

$$\begin{aligned} \|Ty_1 - Ty_2\|_{L^1_q[a,a_1]} &= \|I^{\nu}_{q,a+} [f(t, y_1(t)) - f(t, y_2(t))]\|_{L^1_q[a,a_1]} \\ &\leq A\|I^{\nu}_{q,a+} [y_1 - y_2|\,\|_{L^1_q[a,a_1]} \\ &\leq \omega_1 \|y_1 - y_2\|_{L^1_q[a,a_1]}, \end{aligned}$$

which completes the proof of our first step. The space $L^1_q[a, a_1]$ is the Banach space and according to the Banach's fixed point Theorem (see [24, Subsection 8.1]), there exists a unique solution $\widetilde{y} \in L^1_q[a, a_1]$ such that $T\widetilde{y} = \widetilde{y}'$ for $t \in [a, a_1]$.

Therefore, the solution $v \in L^1_q[a, a_1]$ is obtained from a limit of the convergent sequence $\{(T^i \widetilde{y}_0)\}$:

$$\lim_{i \to 0} \|T^i \widetilde{y}_0 - \widetilde{y}\|_{L^1_q[a,a_1]} = 0, \tag{3.15}$$

where y_0 is any function in $L^1_q[a, a_1]$.

If at least one $\xi_k \neq 0$ in the initial condition (3.4), then we deduce that $\widetilde{y}_0 = y_0$. From (3.14), the sequence $\{(T^i \widetilde{y}_0)(x)\}$ can be written as

$$\left(T^i \widetilde{y}_0\right)(x) = y_0 + \left[I^{\alpha}_{q,a+} f(\cdot, T^{i-1} \widetilde{y}_0(\cdot))\right](x),$$

for $i \in \mathbb{N}$. Let $y_i = T^i \widetilde{y}_0$. If we write $y_i(x) = y_0 + \left(I^{\alpha}_{q,a+} f(\cdot, y_{i-1}(\cdot))\right)(x)$, then it is clear that

$$\lim_{i \to 0} \|y_i - \widetilde{y}\|_{L^1_q[a,a_1]} = 0.$$

Hence, we actually used the method of successive approximations to get a unique solution y' to the integral equation (3.5) on $[a, a_1]$.

Next, we consider the interval $[a_1, a_2]$ and $a_2 = a_1 + h_1$, $a_2 < b$, such that

$$\omega_2 := AK \frac{(a_2 - q a_1)^{\alpha}_q}{\Gamma_q(\alpha + 1)} < 1,$$

for $h_1 > 0$. Then (3.5) can be rewritten as

$$y(x) = y_0(x) + \left(I^{\alpha}_{q,a+} f(\cdot, y(\cdot))\right)(a_1) + \left(I^{\alpha}_{q,a_1+} f(\cdot, y(\cdot))\right)(x).$$

Since the function y is uniquely given on $[a, a_1]$, the first integral can be considered as the known function, and we can rewrite the last equation as

$$(Ty)(x) := y_{10} + \left(I^{\alpha}_{q,a_1+} f(\cdot, y(\cdot))\right)(x),$$

where $y_{10} := y_0(x) + \left(I^{\alpha}_{q,a+} f(\cdot, y(\cdot))\right)(a_1)$. We note that $y_{10} \in L^1_q[a_1, a_2]$.

By using the same arguments as above, we derive that there exists a unique solution $y' \in L^1_q[a_1, a_2]$ to the equation (3.5) on the interval $[a_1, a_2]$. We get the next interval $[a_2, a_3]$, where $a_3 = a_2 + h_2$ such that $a_3 < b$ $(h_2 > 0)$ and repeating the above

process, we obtain a unique solution $y' \in L_q^1[a,b]$ to the equation (3.5) and hence to the Cauchy-type problem (3.3)-(3.4).

Finally, we must show that such a solution $y \in L^1_{\alpha,\beta,\mu,q}[a,b]$ is unique. According to the above proof, the solution $y \in L_q^1[a,b]$ is a limit of the sequence $\{y_i\} \in L_q^1[a,b]$:

$$\lim_{i\to\infty} \|y_i - y\|_{L_q^1[a,b]} = 0, \tag{3.16}$$

with the choice of certain y_i on each $[a,a_1],[a_1,a_2],\ldots,[a_{L-1},b]$, for $L \in \mathbb{N}$. Since

$$\begin{aligned} \|D^{\alpha,\beta}_{q,a+}y_i - D^{(\alpha,\beta)\mu}_{q,a+}y\|_{L_q^1[a,b]} &= \|f(\cdot,y_i(\cdot)) - f(\cdot,y(\cdot))\|_{L_q^1[a,b]} \\ &\leq A\|y_i - y\|_{L_q^1[a,b]}, \end{aligned}$$

by (3.16), we get that

$$\lim_{i\to\infty} \|D^{(\alpha,\beta)\mu}_{q,a+}y_i - D^{(\alpha,\beta)\mu}_{q,a+}y\|_{L_q^1[a,b]} = 0,$$

and hence $D^{\alpha,\beta}_{q,a+} \in L_q^1[a,b]$. This completes the proof. □

4. Miscellaneous Examples

In Section 3, we have introduced sufficient conditions for the Cauchy-type problem to have a unique solution in some subspaces of the space of q-integrable functions. In fact, in the classical case, these conditions show that the q-difference equations of fractional order also can have integrable solutions, provided that their right-hand sides are q-integrable. However, these conditions are not sufficient. Below we present two examples and discuss these examples in connection with the results obtained in Section 3 [23, See examples 3.1-3.2 for fractional cases].

Let $\gamma = \beta + \mu(n-\beta)$ and $\nu = \beta + \mu(\alpha - \beta)$ for $n-1 < \alpha,\beta \leq n, n \in \mathbb{N}$ and $0 \leq \mu \leq 1$.

Example 4.1. We assume that $I^{\gamma}_{q,qa+}y \in AC_q^n[qa,b]$ and $\lambda,\delta \in \mathbb{R}^+$. Then the q-differential equation

$$\left(D^{(\alpha,\beta)\mu}_{q,qa+}\right)(x) = \lambda \frac{(x - q^{-\nu-\delta+1}a)_q^{\nu+\delta}}{(x-q^{-2\nu-\delta+1}a)_q^{\nu}} y^2(x), \tag{4.1}$$

with the initial condition

$$\lim_{x\to qa+}\left(D_q^k I^{(1-\mu)(n-\beta)}_{q,qa+}y\right)(x) = 0 \tag{4.2}$$

has a unique solution in the following form:

$$y(x) = \frac{1}{\lambda}\frac{\Gamma_q(1-\alpha-\gamma)}{\Gamma_q(1-2\alpha-\gamma)}(x-qa)_q^{-\nu-\delta}, \tag{4.3}$$

for $-2\nu-\delta+1>0$, where $(x-a)_q^{\alpha} = x^{\alpha}(a/x;q)_{\alpha}$.

Indeed, we consider

$$f(x,y(x)) := \lambda \frac{(x-q^{-\alpha-\delta+1}a)_q^{\alpha+\delta}}{(x-q^{-2\alpha-\delta+1}a)_q^{\alpha}} y^2(x),$$

and assume that $I^{\gamma}_{q,a+}y \in AC^n_q[qa,b]$ and $G := \{qa \le x \le b; |y(x)| < M < \infty\}$. For $-2\nu-\delta+1>0$, $x\in[qa,b]$ and $y_1,y_2\in G$,

$$\begin{aligned}
|f(x,y_1(x))-f(x,y_2(x))| &= \lambda\left|\frac{(x-q^{-\nu-\delta}a)^{\nu+\delta}_q}{(x-q^{-2\nu-\delta+1}a)^{\nu}_q}y_1^2(x)\right.\\
&- \left.\frac{(x-q^{-\nu-\delta})^{\nu+\delta}_q}{(x-q^{-2\nu-\delta+1}a)^{\nu}_q}y_2^2(x)\right|\\
&\le \lambda\frac{(b-q^{-\nu-\delta+1}a)^{\nu+\delta}_q}{(a-q^{-2\nu-\delta+1}a)^{\nu}_q}\,|y_1(x)+y_2(x)|\,|y_1(x)-y_2(x)|\\
&\le \lambda\frac{(b-q^{-\nu-\delta+1}a)^{\nu+\delta}_q}{(a-q^{-2\nu-\delta+1}a)^{\nu}_q}2M\,|y_1(x)-y_2(x)|\,,
\end{aligned}$$

which proves that f satisfies the Lipschitz condition (3.12) in Theorem 3.4.

Therefore,

$$\begin{aligned}
\int_{qa}^{b}|f(x,y(x))|\,d_qx &= \lambda\int_{qa}^{b}\frac{(x-q^{-\nu-\delta+1}a)^{\nu+\delta}_q}{(x-q^{-2\nu-\delta+1}a)^{\nu}_q}\,|y(x)|^2\,d_qx\\
&\le \lambda\frac{(b-q^{-\nu-\delta+1}a)^{\nu+\delta}_q}{(a-q^{-2\nu-\delta+1}a)^{\nu}_q}M^2,
\end{aligned}$$

which means that $f(x,y(x))\in L^1_q[qa,b]$. We can conclude that, according to Theorem 3.3, the equation has a unique solution in $L^1_q[qa,b]$. Moreover, by applying the operator $D^{(\alpha,\beta)\mu}_{q,qa+}$ (see 3.1)and using the Remark 3.2 and (2.4), (2.5), we get that

$$\begin{aligned}
D^{(\alpha,\beta)\mu}_{q,qa+}(x-qa)^{-\nu-\delta}_q &= I^{\gamma-\nu}_{q,qa+}D^{\gamma}_{q,qa+}(x-qa)^{-\nu-\delta}_q\\
&= \frac{\Gamma_q(1-\nu-\delta)}{\Gamma_q(1-\nu-\delta-\gamma)}\left[I^{\gamma-\nu}_{q,qa+}(x-qa)^{-\nu-\delta-\gamma}_q\right]\\
&= \frac{\Gamma_q(1-\nu-\delta)}{\Gamma_q(1-\delta-2\nu)}(x-qa)^{-2\nu-\delta}_q
\end{aligned}$$

for $2\nu+\delta<1$. Hence,

$$D^{(\alpha,\beta)\mu}_{q,qa+}[y(x)] = \frac{1}{\lambda}\left[\frac{\Gamma_q(1-\nu-\delta)}{\Gamma_q(1-\delta-2\nu)}\right]^2(x-qa)^{-2\nu-\delta}_q\,. \tag{4.4}$$

By using the well-known formulas (see [11, formula (3.7)]):

$$(x-q^{-\nu-\delta+1}a)^{\nu+\delta}_q(x-qa)^{-\nu-\delta}_q = 1$$

and

$$(x-qa)^{-\nu-\delta}_q = (x-qa)^{-2\nu-\delta}_q(x-q^{-2\nu-\delta}a)^{\nu}_q,$$

we have that

$$\begin{aligned} f(x,y(x)) &= \lambda\frac{(x-q^{-\nu-\delta+1}a)_q^{\nu+\delta}}{(x-q^{-2\nu-\delta+1}a)_q^{\nu}}y^2(x) \\ &= \frac{1}{\lambda}\left[\frac{\Gamma_q(1-\nu-\delta)}{\Gamma_q(1-2\nu-\delta)}\right]^2 \times \frac{(x-q^{-\nu-\delta+1}a)_q^{\nu+\delta}}{(x-q^{-2\alpha-\gamma+1}a)_q^{\alpha}}\left[(x-qa)_q^{-\nu-\delta}\right]^2 \\ &= \frac{1}{\lambda}\left[\frac{\Gamma_q(1-\nu-\delta)}{\Gamma_q(1-2\nu-\delta)}\right]^2 (x-qa)_q^{-2\nu-\delta} = D_{q,qa+}^{(\alpha,\beta)\mu}[y(x)]\,. \end{aligned} \tag{4.5}$$

Now, by combining (4.4) and using (4.5) and $\lim_{x\to qa}(x-qa)_q^{-\nu-\delta}=0$ for $x\in[qa,b]$, we see that $y(x)$ defined by (4.3) satisfies (4.1)-(4.2). The proof of our claim is complete.

Example 4.2. Let $\delta,\lambda\in\mathbb{R}^+$. Then the q-differential equation

$$\left(D_{q,a+}^{(\alpha,\beta)\mu}y\right)(x) = \lambda\left[\frac{(x-qa)_q^{2\nu+2\delta}}{(x-q^{\nu+2\delta+1}a)_q^{\nu}}\right]^{\frac{1}{2}}[y(x)]^{\frac{1}{2}}, \tag{4.6}$$

with the initial condition

$$\lim_{x\to qa+}\left(D_q^k I_{q,qa+}^{(1-\mu)(n-\beta)}y\right)(x) = 0 \tag{4.7}$$

has a unique solution in the following form:

$$y(x)=\left[\lambda\frac{\Gamma_q(\nu+2\delta+1)}{\Gamma_q(2\nu+2\delta+1)}\right]^2(x-qa)_q^{2\nu+2\delta}. \tag{4.8}$$

Since $(x-qa)_q^{2\nu+2\delta}=(x-qa)_q^{\nu+2\delta}\left(x-q^{\nu+2\delta}a\right)_q^{\nu}$ we get

$$\begin{aligned} f(x,y(x)) &:= \lambda\frac{\left[(x-qa)_q^{2\nu+2\delta}\right]^{\frac{1}{2}}}{(x-q^{\nu+2\delta+1}a)_q^{\nu}}[y(x)]^{\frac{1}{2}} \\ &= \lambda^2\frac{\Gamma_q(\nu+2\delta+1)}{\Gamma_q(2\nu+2\delta+1)}\left[\frac{(x-qa)_q^{2\nu+2\delta}}{(x-q^{\nu+2\delta+1}a)_q^{\nu}}\right]^{\frac{1}{2}}\left[(x-qa)_q^{2\nu+2\delta}\right]^{\frac{1}{2}} \\ &= \lambda^2\frac{\Gamma_q(\nu+2\delta+1)}{\Gamma_q(2\nu+2\delta+1)}(x-qa)_q^{\nu+2\delta}. \end{aligned} \tag{4.9}$$

We assume that $I_{q,a+}^{\gamma}y\in AC_q^n[a,b]$ and $G:=\{a\le x\le b; M_1<y(x)<M_2, 0<M_1,M_2\}$. Then, for $x\in[a,b]$ and $y_1,y_2\in G$ we obtain

$$\begin{aligned} |f(x,y_1(x))-f(x,y_2(x))| &\le \lambda\frac{\left[(b-qa)_q^{2\nu+2\delta}\right]^{\frac{1}{2}}}{(a-q^{\nu+2\delta+1}a)_q^{\nu}}\left|y_1^{\frac{1}{2}}(x)-y_2^{\frac{1}{2}}(x)\right| \\ &= \lambda\frac{\left[(b-qa)_q^{2\nu+2\delta}\right]^{\frac{1}{2}}}{(a-q^{\nu+2\delta+1}a)_q^{\nu}}\frac{|y_1(x)-y_2(x)|}{y_1^{\frac{1}{2}}(x)+y_2^{\frac{1}{2}}(x)} \\ &\le \frac{\lambda}{2M_1}\frac{\left[(b-qa)_q^{2\nu+2\delta}\right]^{\frac{1}{2}}}{(a-q^{\nu+2\delta+1}a)_q^{\nu}}|y_1(x)-y_2(x)|\,, \end{aligned}$$

which proves that f satisfies the Lipschitz condition (3.6) in Theorem 3.4.

Moreover,

$$\int_{qa}^{b} |f(x,y(x))|\, d_qx \leq \lambda \frac{\left[(b-qa)_q^{2\nu+2\delta}\right]^{\frac{1}{2}}}{(a-q^{\nu+2\delta+1}a)_q^{\delta}} M^{\frac{1}{2}},$$

which means that $f(x,y(x)) \in L_q^1[qa,b]$. We can conclude that according to the Theorem 3.3, the equation has a unique solution in $L_q^1[qa,b]$. Moreover, by applying the operator $D_{q,qa+}^{(\alpha,\beta)\mu}$ (see 3.1) to (4.8) and using Remark 3.2 and (2.4), (2.5), we get that

$$\begin{aligned} D_{q,qa+}^{(\alpha,\beta)\mu}\,(x-qa)_q^{2\nu+2\delta} &= I_{q,qa+}^{\gamma-\nu} D_{q,qa+}^{\gamma}(x-qa)_q^{2\nu+2\delta} \\ &= \frac{\Gamma_q(1+2\nu+2\delta)}{\Gamma_q(1+2\nu+2\delta-\gamma)}\left[I_{q,qa+}^{\gamma-\nu}(x-qa)_q^{2\nu+2\delta-\gamma}\right] \\ &= \frac{\Gamma_q(1+2\nu+2\delta)}{\Gamma_q(1+\nu+2\delta)}(x-qa)_q^{\nu+2\delta} \end{aligned}$$

for $2\nu+\delta<1$. Hence, by 4.9

$$D_{q,qa+}^{(\alpha,\beta)\mu}\,[y(x)] = \lambda^2 \frac{\Gamma_q(1+2\nu+2\delta)}{\Gamma_q(1+\nu+2\delta)}(x-qa)_q^{\nu+2\delta} = f(x,y(x)).$$

Now, by combining (4.4) and using that (4.5) and $\lim_{x\to qa}(x-qa)_q^{-\nu-\delta} = 0$ for $x\in[qa,b]$, we see that $y(x)$ defined by (4.3) satisfies (4.6)-(4.7). The proof of our statement is complete.

5. A linear Cauchy-type q-fractional problem

Let $\gamma=\beta+\mu(n-\beta)$ and $\nu=\beta+\mu(\alpha-\beta)$ for $n-1<\alpha,\beta\leq n, n\in\mathbb{N}$ and $0\leq\mu\leq 1$.

The (Mittag-Leffler) q-function $E_{\alpha,\beta}\,(z;q)$ is defined by ([28])

$$E_{\alpha,\beta}\,[\lambda x^{\gamma}(a/x;q)_{\gamma};q] = \sum_{k=0}^{\infty} \frac{\lambda^k x^{k\gamma}(a/x;q)_{k\gamma}}{\Gamma_q(\alpha k+\beta)}. \tag{5.1}$$

We study the following Cauchy type q-fractional problem

$$\left(D_{q,a+}^{(\alpha,\beta)\mu} y\right)(x) - \lambda y(x) = f(x), \tag{5.2}$$

$$\lim_{x\to a+}\left(D_q^k I_{q,a+}^{(1-\mu)(n-\beta)} y\right)(x) = \xi_k, \quad \xi_k\in\mathbb{R}, k=0,1,2,\ldots n-1. \tag{5.3}$$

Theorem 5.1. *Let $\gamma=\beta+\mu(n-\beta)$ and $\nu=\beta+\mu(\alpha-\beta)$ for $n-1<\alpha,\beta\leq n, n\in\mathbb{N}$ and $0\leq\mu\leq 1$. We assume that $\lambda\in\mathbb{R}$ is such that*

$$|\lambda| b^{\nu}(1-q)^{\nu}<1,$$

and $f\in L_q^1[a,b]$. Then the Cauchy problem (5.2)-(5.3) has a unique solution $y\in L_{\alpha,\beta,\mu,q}^1[a,b]$ and this solution is given by

$$\begin{aligned} y(x) &= \sum_{k=1}^{n} b_k x^{\nu-k} E_{\nu,\nu-k+1}\left[\lambda x^{\nu}; q\right] \\ &+ \int_0^x x^{\nu-1} \left(qt/x; q\right)_{\nu-1} E_{\nu,\nu}\left[\lambda x^{\nu} (q^{\nu} t/x; q)_{\nu}; q\right] f(t) d_q t. \end{aligned} \tag{5.4}$$

Proof. First, we solve the Volterra q-integral equation (3.5) for

$$f(x, y(x)) := \lambda y(x) + f(x).$$

For this aim, we apply the method of successive approximations by setting

$$y_0(x) = \sum_{k=1}^{n} \frac{\xi_k}{\Gamma_q(\gamma - k + 1)} x^{\gamma-k} (a/x; q)_{\gamma-k}$$

and

$$\begin{aligned} y_i(x) &= y_0(x) + \frac{\lambda x^{\nu-1}}{\Gamma_q(\nu)} \int_0^x (qt/x; q)_{\nu-1} y_{i-1}(t) d_q t \\ &+ \frac{x^{\nu-1}}{\Gamma_q(\nu)} \int_0^x (qt/x; q)_q^{\nu-1} f(t) d_q t. \end{aligned} \tag{5.5}$$

Using (2.3) and (5.5) we find $y_1(x)$:

$$y_1(x) = y_0(x) + \lambda \left(I_{q,0+}^{\nu} y_0\right)(x) + \left(I_{q,0+}^{\nu} f\right)(x),$$

that is,

$$\begin{aligned} y_1(x) &= \sum_{k=1}^{n} \frac{b_k}{\Gamma_q(\nu - k + 1)} x^{\nu-k} + \lambda \sum_{k=1}^{n} \frac{b_k}{\Gamma_q(\nu - k + 1)} \left(I_{q,0+}^{\nu} t^{\nu-k}\right)(x) + \left(I_{q,0+}^{\nu} f\right)(x) \\ &= \sum_{k=1}^{n} \frac{b_k}{\Gamma_q(\nu - k + 1)} x^{\nu-k} + \lambda \sum_{k=1}^{n} \frac{b_k x^{2\nu-k}}{\Gamma_q(2\nu - k + 1)} + \left(I_{q,0+}^{\nu} f\right)(x) \\ &= \sum_{k=1}^{n} b_k \sum_{m=1}^{2} \frac{\lambda^{m-1} x^{m\nu-k}}{\Gamma_q(\nu m - k + 1)} + \left(I_{q,0+}^{\nu} f\right)(x). \end{aligned} \tag{5.6}$$

Similarly, using (2.3), (2.4), (2.7) and (5.6) we have for $y_2(x)$ that

$$\begin{aligned} y_2(x) &= y_0(x) + \lambda \left(I^{\nu}_{q,0+} y_1\right)(x) + \left(I^{\nu}_{q,0+} f\right)(x) \\ &= \sum_{k=1}^{n} \frac{b_k}{\Gamma_q(\nu-k+1)} x^{\nu-k} \\ &+ \frac{\lambda}{\Gamma_q(\nu)} \sum_{k=1}^{n} b_k \sum_{m=1}^{2} \frac{\lambda^{m-1}}{\Gamma_q(\nu m-k+1)} \left(I^{\nu}_{q,0+} t^{m\nu-k}\right)(x) \\ &+ \lambda \left(I^{\nu}_{q,0+} I^{\nu}_{q,0+} f(t)\right)(x) + \left(I^{\nu}_{q,0+} f\right)(x) \\ &= \sum_{k=1}^{n} \frac{b_k}{\Gamma_q(\nu-k+1)} x^{\nu-k} + \lambda \sum_{k=1}^{n} b_k \sum_{m=1}^{2} \frac{\lambda^{m-1}}{\Gamma_q(\nu(m+1)-k+1)} x^{\nu(m+1)-k} \\ &+ \lambda \left(I^{2\nu}_{q,0+} f(t)\right)(x) + \left(I^{\nu}_{q,0+} f\right)(x) \end{aligned}$$

$$\begin{aligned} &= \sum_{k=1}^{n} \frac{b_k}{\Gamma_q(\nu-k+1)} x^{\nu-k} + \lambda \sum_{k=1}^{n} b_k \sum_{m=1}^{2} \frac{\lambda^{m-1}}{\Gamma_q(\nu(m+1)-k+1)} x^{\nu(m+1)-k} \\ &+ \frac{\lambda x^{2\nu-1}}{\Gamma(2\nu)} \int_0^x f(t) (qt/x;q)_{2\nu-1} \, d_q t + \left(I^{\nu}_{q,0+} f\right)(x). \end{aligned}$$

Thus,

$$\begin{aligned} y_2(x) &= \sum_{k=1}^{n} b_k \sum_{m=1}^{3} \frac{\lambda^{m-1} x^{\nu m-k}}{\Gamma_q(\nu m-k+1)} \\ &+ \int_0^x \left[\sum_{m=1}^{2} \frac{\lambda^{m-1} x^{\nu m-1} (qt/x;q)_{\nu m-1}}{\Gamma_q(\nu m)} \right] f(t) d_q t. \end{aligned}$$

Continuing this process, we derive the following relation for $y_i(x)$:

$$\begin{aligned} y_i(x) &= \sum_{k=1}^{n} b_k \sum_{m=1}^{i+1} \frac{\lambda^{m-1} x^{\nu m-k}}{\Gamma_q(\nu m-k+1)} \\ &+ \int_0^x \left[\sum_{m=1}^{i} \frac{\lambda^{m-1} x^{\nu m-1} (qt/x;q)_{\nu m-1}}{\Gamma_q(\nu m)} \right] f(t) d_q t \\ &= \sum_{k=1}^{n} b_k \sum_{m=0}^{i} \frac{\lambda^{m} x^{\nu(m+1)-k}}{\Gamma_q(\nu(m+1)-k+1)} \\ &+ \int_0^x \left[\sum_{m=0}^{i-1} \frac{\lambda^{m} x^{\nu(m+1)-1} (qt/x;q)_{\nu(m+1)-1}}{\Gamma_q(\nu(m+1))} \right] f(t) d_q t. \end{aligned}$$

From (2.1) and (2.2) its follows that

$$\begin{aligned}\frac{\lambda^m x^{\nu m}(qt/x;q)_{\nu m}}{\Gamma_q(\nu m+\nu)} &= [\lambda x^\nu(1-q)^\nu]^m \frac{(q^{\nu(m+1)};q)_\infty}{(q^{\nu m+1}t/x;q)_\infty}(1-q)^{\nu-1}\\ &\leq [\lambda x^\nu(1-q)^\nu]^m \frac{(q^{\nu(m+1)};q)_\infty}{(q^{\nu m+1};q)_\infty}(1-q)^{-1}\\ &\leq [\lambda b^\nu(1-q)^\nu]^m (1-q)^{-1}.\end{aligned}$$

Thus,

$$\left|\sum_{m=0}^{\infty}\frac{\lambda^m x^{\nu m}(qt/x;q)_{\nu m}}{\Gamma_q(\nu m+\nu)}\right| \leq \frac{1}{1-q}\sum_{m=0}^{\infty}[|\lambda|b^\nu(1-q)^\nu]^m \leq \frac{1}{|\lambda|b^\nu(1-q)^{\nu+1}},$$

which means that this is absolutely and uniformly convergent for $\lambda b^\nu(1-q)^\nu < 1$. Taking the limit as $i \to \infty$ and using the well-known formulas (see [11, formula (3.7)]), we obtain the following explicit solution $y(x)$ to the q-integral equation (5.4):

$$\begin{aligned}y(x) &= \sum_{k=1}^{n} b_k x^{\nu-k}\sum_{m=0}^{\infty}\frac{\lambda^m x^{\nu m}}{\Gamma_q(\nu m+\nu-k+1)}\\ &+ \int_0^x x^{\nu-1}(qt/x;q)_{\nu-1}\left[\sum_{m=0}^{\infty}\frac{\lambda^m x^{\nu m}(qt/x;q)_{\nu m}}{\Gamma_q(\nu m+\nu)}\right] f(t)d_qt.\end{aligned}$$

On the basis of Theorem 3.4 and (5.1), the last expression represents the explicit solution to the Volterra q-integral equation (5.4) and hence to the Cauchy-type problem (5.2)-(5.3). □

6. Conflict of Interests

The authors declare that they have no conflict of interest.

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